

The Carry Independence Conjecture in Two-Clock Form: the LTE round chain, exact Haar laws, and the entropy deficit

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Abstract

The Carry Independence Conjecture (CIC) of arXiv:2603.11066 — every odd $n_0 > 1$ eventually takes a Syracuse letter $b \geq 3$ — is the weakest nontrivial case of our two-clock calibration conjecture (it asserts the handoff parity cannot equal 1 forever). We attack it with the lifting-the-exponent machinery. Results: (1) convergent orbits satisfy the CIC automatically (every entry into 1 uses a letter $b \geq 4$), so the CIC concerns only divergent $\{1, 2\}$ -orbits; (2) a $\{1, 2\}$ -orbit is exactly an iterated chain of odd integers in which each survival event is the LTE reading $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$ — alignment of the orbit’s fresh high bits to a *rotating 2-adic target* — with death iff the reading is odd; (3) under Haar measure the round process is *exactly* i.i.d.: survival probability $\frac{1}{3}$ per round, geometric laws on every fiber (proved via the census lemma; the freshness slope is $2 \cdot 3^{\#\text{letters}}$), and surviving rounds drift *upward* at $+0.6166$ bits per round in the mean — the apparent anti-correlations in finite enumerations are truncation artifacts; (4) Mersenne points $2^g - 1$ survive their first round iff g is odd and $v_2(g - 1)$ is even (pure LTE, deterministic); (5) the $\{1, 2\}$ alphabet has an *entropy deficit*: confined-word growth is bounded by $2^{H(2-\log_2 3)} = 1.9712 < 2$ at *every* strip width, so the narrow-band dwell machinery applies to CIC-violators with no width cutoff — the allowed region of the (D, ε) phase diagram is empty for them; (6) empirically, the least integer surviving D letters grows like $2^{D/2.2}$. The remaining wall is a single scalar parity sequence read from provably fresh bits.

1 Reduction to divergent $\{1, 2\}$ -orbits

Lemma 1 (Convergent orbits satisfy the CIC). *If the Syracuse orbit of $n_0 > 1$ reaches 1, it takes a letter $b \geq 4$ at its last step before 1.*

Proof. $S(y) = 1$ means $3y + 1 = 2^b$, i.e. $y = (4^k - 1)/3$ with $b = 2k$ (only even b give integers). $k = 1$ is $y = 1$; every entry from $y \neq 1$ has $b = 2k \geq 4$. □

Corollary 1. *A CIC-violator is either a nontrivial positive cycle with all letters in $\{1, 2\}$ — excluded in all computationally verified ranges, and period-pinned by the Baker mechanism of the narrow-band note — or a divergent orbit with all letters in $\{1, 2\}$. The CIC is henceforth the statement that no divergent $\{1, 2\}$ -orbit exists.*

2 The round chain and its LTE normal form

Theorem 1 (Round chain). *Let $x \equiv 3 \pmod{4}$ be a 1-run start, $g = v_2(x + 1)$, $\ell = g - 1$, $m = (x + 1)/2^g$ odd. Set $v = v_2(3^\ell m - 1)$. Then the next letters of x are exactly $1^\ell 2^{\lfloor v/2 \rfloor}$ followed by: a*

letter $b \geq 3$ if v is odd (death), or a new 1-run if v is even (survival), in which case, with $r = v/2$ and $u = (3^\ell m - 1)/2^v$,

$$m' = \frac{3^r u + 1}{2^{v_2(3^r u + 1)}}, \quad \ell' = v_2(3^r u + 1),$$

are the parameters of the next round. Moreover the reading has the LTE normal form

$$v = v_2(3^\ell m - 1) = v_2(m - 3^{-\ell}) :$$

each survival event is an alignment reading of the fresh multiplier m against the rotating 2-adic target $3^{-\ell} \in \mathbb{Z}_2^\times$. (Machine-verified: 5000 rounds, letters, death criterion and (g', m') all exact; LTE form on 2000 random (ℓ, m) .)

Proof. The -1 clock gives the run 1^ℓ ending at y with $y - 1 = 2(3^\ell m - 1)$, so $h = v_2(y - 1) = 1 + v$; the $+1$ clock gives the 2-run of length $\lfloor (h - 1)/2 \rfloor = \lfloor v/2 \rfloor$ with parity handoff: h odd (v even) hands to a 1-run, h even (v odd) forces $b \geq 3$. The terminal point is z with $z + 1 = 2(3^r u + 1)$, giving (g', m') . The normal form is $v_2(3^\ell) = 0$ plus invertibility of 3 in $\mathbb{Z}_2$. $\square$

Remark. The CIC is thus: no odd integer can feed the chain even readings forever. The targets $3^{-\ell}$ rotate through the index-4 subgroup $\langle 3 \rangle \subset (\mathbb{Z}/2^j)^\times$, and at scale j the target depends only on $\ell \bmod 2^{j-2}$ — the lifting-the-exponent structure.

3 Exact Haar laws: the survival coin is i.i.d.

Theorem 2 (Freshness; exact i.i.d. laws). *Condition on the full data of rounds $1, \dots, i$ (all letters and all read bits). Then the round- $(i+1)$ reading is exactly geometric: $\mathbb{P}[v = k] = 2^{-k}$ ($k \geq 1$), hence*

$$\mathbb{P}[\text{survive round}] = \sum_{k \geq 2 \text{ even}} 2^{-k} = \frac{1}{3}$$

independently across rounds; given survival, $\mathbb{P}[r = t] = 3 \cdot 4^{-t}$ ($\mathbb{E}[r] = \frac{4}{3}$) and the next depth is $\mathbb{P}[g' = j] = 2^{-(j-1)}$ ($\mathbb{E}[\ell'] = 2$).

Proof sketch. Write the unread part of m as $m = a + 2^w s$ with s the free (Haar) parameter. Through the round, every derived quantity is affine in s with slope of the exact form $2 \cdot 3^B \cdot s$ (B = letters consumed): $u = u_0 + 2 \cdot 3^\ell s$, $3^r u + 1 = T_0 + 2 \cdot 3^{\ell+r} s$, and after extracting $2^{g'}$ and conditioning, $m' = m'_0 + 2 \cdot 3^B s'$ — the same form, renormalized. The census lemma of the ghost-clock note (slope valuation $\gamma = 1 \leq \beta$) then gives the exact geometric law of each reading on each fiber, by induction over rounds. (This is the orbit-parameterization slope $2 \cdot 3^D$ of arXiv:2603.11066 composed with the census lemma; it generalizes that manuscript's Carry Contamination Theorem from mod-8 equidistribution of one round to exact i.i.d. laws of the whole process.) $\square$

Verification. Exact enumeration of all 1-run-start residue classes modulo 2^{24} , counting only classes whose round outcomes are class-determined: round-1 survival 1398024/4194144 = 0.333328, round-2 0.332689. Deeper rounds show 0.322, 0.272, 0.125 — this is *truncation bias*, not anti-correlation: the determinedness filter caps the current reading at the remaining bit budget, and a geometric law truncated at an odd cutoff has even-mass below $\frac{1}{3}$ (e.g. $2/7$ at cutoff 3). Any empirical “two-round anti-correlation” at finite modulus has this origin; under Haar the coin is exactly fair-thirds and i.i.d.

Corollary 2 (Survivors ascend). *Conditioned on survival, the mean drift per round is*

$$(\log_2 3 - 1) \mathbb{E}[\ell] - (2 - \log_2 3) \mathbb{E}[r] = 2(0.58496) - \frac{4}{3}(0.41504) = +0.6166 \text{ bits},$$

i.e. +0.185 bits per letter (verified: +0.6164 over the enumerated survivor classes). Haar-typical points die (density $(1/3)^j$ after j rounds, so $\mathcal{C}_\infty$ is a measure-zero Cantor set), while the surviving fibers drift upward at a definite rate: a positive CIC-violator, if one exists, lives in an exponentially thin set and is statistically forced toward divergence — consistent with Corollary 1.

4 Deterministic sub-families: Mersenne LTE

Theorem 3 (Mersenne first round). *For $x = 2^g - 1$ ($m = 1$, $\ell = g - 1$) the reading is fully deterministic: $v = v_2(3^{g-1} - 1)$ equals 1 if g is even, and $2 + v_2(g - 1)$ if g is odd (LTE). Hence $2^g - 1$ survives its first round iff g is odd and $v_2(g - 1)$ is even. (Verified $g \leq 60$: survivors $g \in \{5, 13, 17, 21, 29, 37, 45, 49, 53\}$, i.e. $g - 1 \equiv 2^{2t} \pmod{2^{2t+1}}$.)*

Remark. *The truncated ghosts are precisely the points whose readings are not fresh — and there LTE decides everything. Freshness (Theorem 2) and LTE determinism (Theorem 3) are the two poles of the problem: the wall is exactly the regime between them.*

5 The entropy deficit of the $\{1, 2\}$ alphabet

Theorem 4 (Entropy deficit). *Let $N_D^{12}(p)$ count D -confined valuation words of length p over the alphabet $\{1, 2\}$, and $\lambda_{12}(D) = \lim N_D^{12}(p)^{1/p}$. Then for every D ,*

$$\theta_{12}(D) = \log_2 \lambda_{12}(D) \leq H(2 - \log_2 3) = 0.97907 < 1,$$

since confinement pins the letter-1 frequency at $2 - \log_2 3$ and binary words at fixed frequency number at most $2^{H(q)p+o(p)}$. Exact quantized counts (rotation-driven recursion, $p = 2000$): $\theta_{12} = 0.521, 0.753, 0.889, 0.955, 0.973$ at $D = 0.8, 1.2, 2, 4, 8$, increasing toward the ceiling.

Corollary 3 (All-width dwell for CIC-violators). *An acyclic $\{1, 2\}$ -orbit dwells at most $C(WY)^{0.980}$ consecutive steps in any band $[Y, WY]$ — including arbitrarily wide bands. (Proof: repetition rigidity plus the pigeonhole of the narrow-band theorem, with N^{12} in place of N ; no D^* cutoff because $\theta_{12} < 1$ uniformly.) In particular, a single excursion of a divergent $\{1, 2\}$ -orbit below height X lasts at most $C X^{0.980}$ steps, and the allowed (red) region of the (D, ε) phase diagram is empty for CIC-violators: the full-alphabet escape hatch at $D > D^* = 1.246$ is closed by the missing letters.*

6 Survival champions

The least $n_0 \equiv 3 \pmod{4}$ surviving D letters (seeds to 2^{26}):

D	4–19	20	28	36	39–44
$\min n_0$	27	4591	31911	517191	687871
$D/\log_2 n_0$	≤ 4.0	1.64	1.87	1.90	2.0–2.3

The ratio $D/\log_2(\min n_0)$ stabilizes near 2: empirically the least survivor of depth D has height about $2^{D/2.2}$. We record:

Conjecture 1 (Champion growth). *There is $c > 0$ with $\min\{n_0 : n_0 \text{ survives } D \text{ letters}\} \geq 2^{cD}$. Granting this, every CIC-violating orbit satisfies $x_n \geq 2^{c'n}$: violators must flee to infinity at a definite exponential rate, closing the CIC for all sub-exponential orbits.*

7 The 3-smooth cascade and the 2-adic logarithm calculus

The Mersenne theorem extends to the full 3-smooth neighborhoods $x = 2^g 3^a \mp 1$ — and the cascade goes *deeper* than one round. All statements below are machine-verified (`search/smooth_cascade.py`; grids $g < 42$, $a < 26$, and the one-parameter family to $n \leq 3000$).

Theorem 5 (Minus family). *For $x = 2^g 3^a - 1$ ($g \geq 2$) the entire round chain depends on (g, a) only through $n = g + a - 1$ (the chain sees m, ℓ only through $3^\ell m = 3^n$). The first reading is $v = v_2(3^n - 1)$, given by LTE: $v = 1$ for n odd, $v = 2 + v_2(n)$ for n even; survival iff $v_2(n)$ is even ≥ 2 , of density $\frac{1}{6}$ over n — the family is ghost-aligned and dies at $2\times$ the Haar rate in round one. The round-2 reading is the 4-term S -unit valuation*

$$v' = v_2(3^{n+r+\ell'} - 3^{r+\ell'} + 2^v 3^{\ell'} - 2^{v+\ell'}) - v - \ell'.$$

Theorem 6 (Plus family). *For $x = 2^g 3^a + 1$ ($g \geq 3$) the $+1$ clock rides $\lfloor (g-1)/2 \rfloor$ blocks ending 3-smooth, with $a' = a + \lfloor (g-1)/2 \rfloor$:*

(i) *g even: terminal $4 \cdot 3^{a'} + 1$, death letter $b = 2 + v_2(3^{a'+1} + 1) \in \{3, 4\}$ by LTE — closed-form CIC death.*

(ii) *g odd: lands at $2 \cdot 3^{a'} + 1$, next depth $g' = 2$ (a' even) or 3 (a' odd). If a' is even the round-2 reading is $v' = v_2(3^{a'+1} + 1) - 1 = 1$: closed-form death. If a' is odd,*

$$v' = v_2(3^{a'+2} + 5) - 2 = v_2(a' + 2 - \text{Log}_3(-5)) :$$

the survival reading is the 2-adic alignment of the exponent to the 2-adic logarithm of the ghost -5 , where $N_0 = \text{Log}_3(-5) = 2556611305511861515 \bmod 2^{64}$ and $v_2(3^N + 5) = 2 + v_2(N - N_0)$ for $N \equiv N_0 \bmod 2$.

Proposition 1 (The S -unit horizon). *Along any 3-smooth-seeded chain, the round- j reading is the 2-adic valuation of a 2–3 S -unit form whose term count grows with j ; two-term forms are LTE (closed), and the collapsed cases reduce to alignments of integer exponents to 2-adic logarithms of ghosts. The deterministic horizon of the cascade is exactly the S -unit term count the machinery can close — the CIC on these families meets the banded S -unit frontier (Target 1) at the level of exponents, where effective p -adic Baker theory (Yu) bounds every reading by $O(\log)$ of the exponent height.*

Cascade statistics (one-parameter family, $n \leq 3000$): survival 0.16633 after round 1 (LTE prediction $\frac{1}{6}$), then conditional survivals 0.34, 0.31, 0.40, 0.19, 0.25 — the re-randomized fair-thirds coin of Theorem 2. The champion is $n = 704$, surviving 6 rounds with readings $[8, 2, 2, 2, 2, 2, 3]$; every $n \leq 3000$ dies by round 7: the CIC holds, with explicit finite certificates, for the entire 3-smooth slice tested. The ghost hierarchy thus lifts one level: at the base, integers align to ghosts $(-1, +1, -5, \dots)$; at the exponent level, exponents align to 2-adic logarithms of ghosts; survival is a parity reading at each level, deterministic exactly where freshness fails.

8 Where the wall now stands

The CIC has been compressed to: *a single scalar parity sequence — the readings $v_2(m_i - 3^{-\ell_i})$ — cannot be even forever along any positive orbit.* The readings are provably i.i.d. fair-thirds coins on Haar fibers (Theorem 2); they are provably deterministic, and lethal in explicit congruence patterns, on the ghost-aligned sub-families (Theorem 3); the counting machinery confines any violator to perpetual fast ascent with empty phase-diagram refuge (Theorem 4, Corollary 2); and the champion data suggest violators must be exponentially large in their survival depth. What is missing is precisely the distributional-to-pointwise step — the same quenched calibration wall, now localized to one coin.

References

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